Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE ADVANCED - 3 | PAPER - 1

JEE 2022

Date	Maximam Marks	Duration
1st August, 2021	198	3 Hours

General Instructions

1. The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **three** sections (Section 1, Section 2 & Section 3).

2. **Section 1** contains **6 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

   Section 2 contains **6 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

   Section 3 contains **6 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

3. For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) : ______________________

Roll Number : ______________________

OMR Bar Code Number : ______________________

Candidate's Signature : ______________ Invigilator's Signature ______________

MARKING SCHEME

SECTION - 1 (Maximum Marks: 18)

- This section contains **SIX (06)** Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If only (all) the correct option(s) is(are) chosen
Zero Mark: 0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks: –1 In all other cases.

SECTION - 2 (Maximum Marks: 24)

- This section consists of **SIX (06)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If only (all) the correct option(s) is(are) chosen
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark: 0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks: –2 In all other cases.

SECTION – 3 (Maximum Marks: 24)

- This section contains **6 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

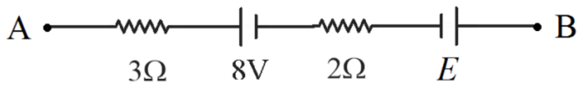
Full Marks: +4 If ONLY the correct Integer value is entered. There is **NO negative marking.**
Zero Marks: 0 In all other cases.

SUBJECT I : PHYSICS 66 MARKS

SECTION 1
SINGLE CORRECT ANSWERS TYPE

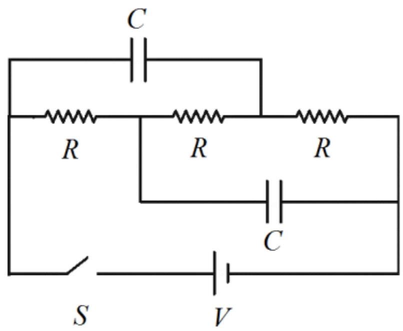
This section contains 06 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A part of a larger circuit is shown in the figure. Both batteries shown have internal resistance $1\,\Omega$ each. If the points A and B are equipotential and a current 2 A flows from A towards B, then the emf E (in Volts) is:



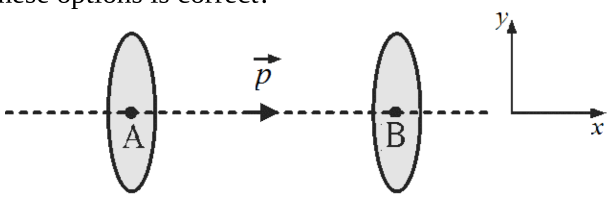
(A) 14 **(B)** 16 **(C)** 18 **(D)** 22

2. In the circuit shown, the battery is ideal, and both capacitors are initially uncharged. If the switch S is closed, the current through the switch immediately afterwards, and the current through it after a long time is respectively:



(A) $\frac{V}{3R}, \frac{3V}{R}$ **(B)** $\frac{3V}{R}, \frac{V}{3R}$ **(C)** $\frac{3V}{R}$, *zero* **(D)** *zero*, $\frac{3V}{R}$

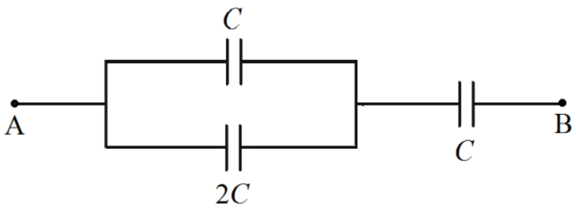
3. A small dipole is fixed at the origin with its moment pointing in the +X direction. Consider two identical circular disc shaped imaginary surfaces with their centres at two points A and B on the X-axis equidistant from the origin, shown in the figure. The surfaces are perpendicular to the X-axis. The electric flux through the surfaces is Φ_A and Φ_B respectively. The sign of flux is taken positive if it crosses a surface towards the +X direction and taken negative if it crosses a surface towards the –X direction. Which of these options is correct?



(A) $\Phi_A > 0$ and $\Phi_B > 0$ **(B)** $\Phi_A < 0$ and $\Phi_B > 0$

(C) $\Phi_A > 0$ and $\Phi_B < 0$ **(D)** $\Phi_A < 0$ and $\Phi_B < 0$

4. In the given circuit, all three capacitors are initially uncharged. If a battery of emf E is connected between the points A and B, the potential difference across the capacitor of capacitance $2C$ at steady state becomes:



(A) $\frac{E}{4}$ **(B)** $\frac{2E}{5}$ **(C)** $\frac{3E}{5}$ **(D)** $\frac{3E}{4}$

5. A non-conducting ball carrying a charge $+Q$ is placed on a smooth horizontal non-conducting table and tied to a spring of spring constant k whose other end is fixed. Initially the spring is unstretched. Now, another identical ball, also carrying charge $+Q$, is slowly moved closer from a very large initial distance directly towards the first ball until it reaches the position initially occupied by the first ball. If the work done in the process is proportional to $Q^M k^N$, then:
(The balls are small enough that the electrostatic force between them can be calculated by assuming that they are point charges)

(A) $M=2, N=\frac{4}{3}$ **(B)** $M=\frac{2}{3}, N=\frac{1}{3}$ **(C)** $M=\frac{2}{3}, N=1$ **(D)** $M=\frac{4}{3}, N=\frac{1}{3}$

6. The region $r<\frac{R}{2}$ of a non-conducting sphere (r denotes the distance from the centre of the sphere) is uncharged, and the region $\frac{R}{2} \le r \le R$ is charged with charge density $\rho(r)=\rho_0\left(\frac{r}{R}\right)$, where ρ_0 is a constant. In the region $\frac{R}{2} \le r \le R$, the electric field $E(r)$ varies as:

(A) $E(r)=\frac{\rho_0}{4\varepsilon_0 R}\left(r^2-\frac{R^4}{4r^2}\right)$ **(B)** $E(r)=\frac{\rho_0}{4\varepsilon_0 R}\left(r^2-\frac{R^4}{16r^2}\right)$

(C) $E(r)=\frac{\rho_0}{16\varepsilon_0 R^2}\left(r^3-\frac{R^4}{4r}\right)$ **(D)** $E(r)=\frac{\rho_0}{16\varepsilon_0 R^2}\left(r^3-\frac{R^4}{16r}\right)$

SPACE FOR ROUGH WORK

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of 06 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

7. Two thin conducting spherical shells of radii R and $3R$ are fixed concentrically, and given total charge $-Q$ and $+Q$ respectively. Which of these options is/are correct?

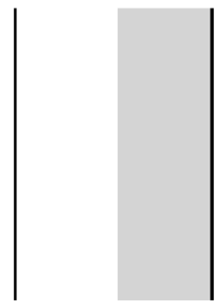
(A) The electric field at a point at a distance $2R$ from the their common centre has magnitude $\dfrac{Q}{16\pi\varepsilon_0 R^2}$ and is directed radially inward

(B) The electric field inside the inner shell is zero

(C) The potential at their common centre is $-\dfrac{Q}{6\pi\varepsilon_0 R}$

(D) The potential of the outer shell is zero

8. The separation between the plates of an uncharged capacitor of capacitance C is d. Initially, there is air (dielectric constant 1) between the plates. A slab of a material of dielectric constant 2 and thickness $\dfrac{d}{2}$ is introduced inside the capacitor as shown. Now, the capacitor is connected across the terminals of a battery of emf V. After the capacitor is completely charged:



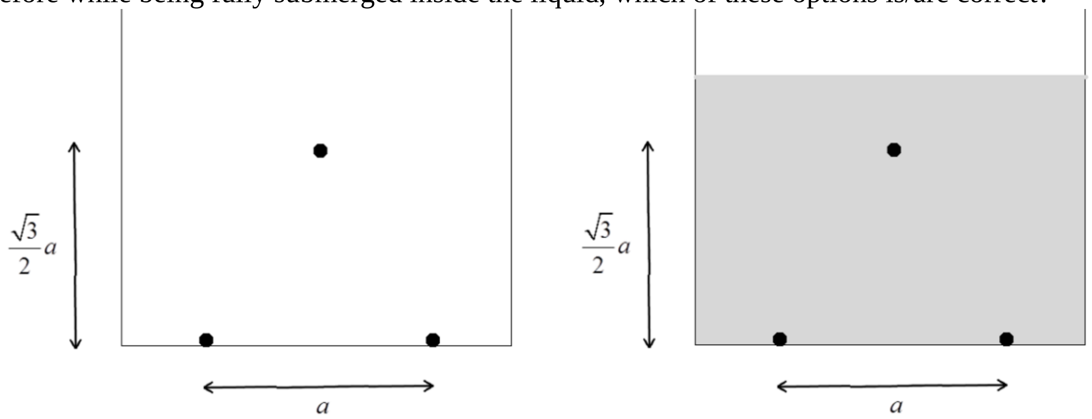
(A) The electric field inside the slab is $\dfrac{4V}{3d}$

(B) The electric field inside the slab is $\dfrac{2V}{3d}$

(C) The potential energy stored in the capacitor is $\dfrac{2}{3}CV^2$

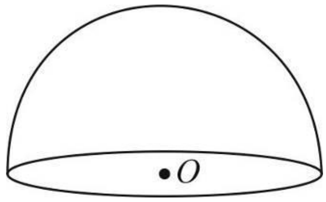
(D) The potential energy stored in the capacitor is $\dfrac{1}{3}CV^2$

9. Two small balls carrying charge $+q$ each are fixed to the bottom of an empty non-conducting vessel, separated by a distance a as shown. A third small ball of density ρ_0, mass m and carrying the same charge $+q$ floats in equilibrium on the perpendicular bisector of the other two balls at a height $\dfrac{\sqrt{3}}{2}a$ above them, with all three balls being in the same vertical plane. Now, the vessel is filled with a liquid of density ρ and dielectric constant κ. If the floating ball remains in equilibrium at the same position as before while being fully submerged inside the liquid, which of these options is/are correct?



(A) $mg = \frac{\sqrt{3}q^2}{16\pi\varepsilon_0 a^2}$ **(B)** $mg = \frac{\sqrt{3}q^2}{4\pi\varepsilon_0 a^2}$ **(C)** $\kappa = \frac{2\rho_0}{\rho_0 + \rho}$ **(D)** $\kappa = \frac{\rho_0}{\rho_0 - \rho}$

10. An insulating hemispherical shell of radius R is fixed and charged uniformly over its curved surface with charge per unit area σ. The flat, circular base of the shell is uncharged. Let O be the centre of the circular base. Which of these options is/are correct?



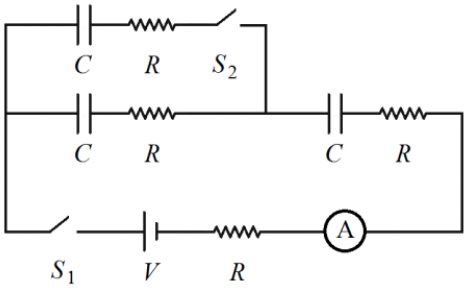
(A) The circular base is an equipotential surface

(B) The electric field at O is zero

(C) The work done in moving a point charge $+q$ from infinity to O is $\frac{\sigma R q}{2\varepsilon_0}$

(D) The electric potential at a point in the same plane as the circular base, and at a distance $2R$ from O, is $\frac{\sigma R}{4\varepsilon_0}$

11. Three identical uncharged capacitors, each of capacitance C, are connected in a circuit with an ideal battery and an ammeter of resistance R as shown. Initially both the switches S1 and S2 are open. Now, S1 is closed. After the current in the ammeter has become negligible, S2 is closed while keeping S1 closed. Which of these options is/are correct?



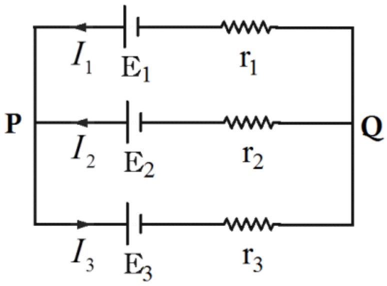
(A) The current in the ammeter immediately after S1 is closed is $\frac{V}{4R}$

(B) The work done by the battery during the time S1 is closed and S2 is open is $\frac{1}{2}CV^2$

(C) The current in the ammeter immediately after S2 is closed is $\frac{V}{14R}$

(D) The work done by the battery over a long time after S2 is closed is $\frac{1}{6}CV^2$

12. Three ideal batteries of emf E_1, E_2 and E_3 and three resistances r_1, r_2 and r_3 are connected as shown. Currents I_1, I_2 and I_3 flow in the three branches along the directions indicated. Which of these options is/are correct? [Given : $E_1 = 4V$, $E_2 = 4V$, $E_3 = 2V$, $r_1 = 1\Omega, r_2 = 2\Omega$, $r_3 = 2\Omega$]



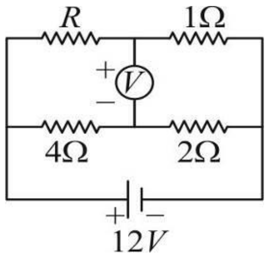
(A) $I_3 = 0.75A$ **(B)** $V_P - V_Q = 4V$ **(C)** $V_P - V_Q = 3.5V$ **(D)** $I_1 = 0.5V$

SECTION 3

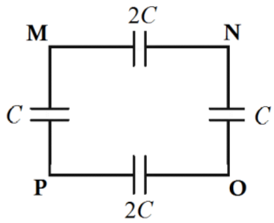
NUMERICAL VALUE TYPE QUESTIONS

This section has 06 Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25)

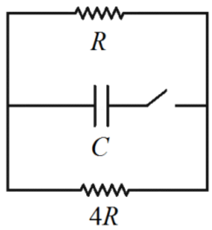
13. The battery and the voltmeter in the given circuit are ideal. The voltmenter reads –1 V. The resistance R is equal to ___________Ω.



14. All capacitors in the network shown are initially uncharged. If a battery of emf E is connected between the points M and O, and the potential difference between the points N and P at steady state is V_{NP}, the ratio $\frac{V_{NP}}{E}$ is _______________.



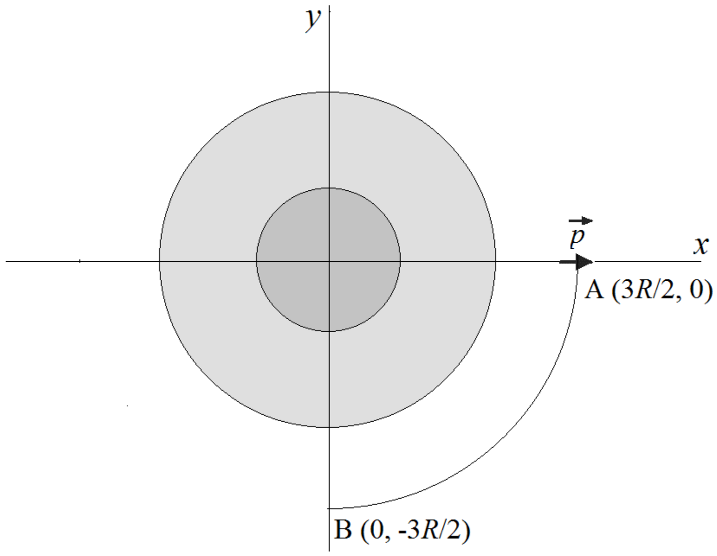
15. A capacitor of capacitance C is charged to a potential difference V_0 and then connected to two resistances R and $4R$ as shown. The total resistance of the branch containing the capacitor is negligible. After the switch is closed, the total heat energy dissipated in the resistance R until the capacitor is completely discharged is $X\left(CV_0^2\right)$. The value of X is ____________.



16. Two thin circular rings A and B, both of radius R, are fixed coaxially with a distance $\sqrt{3}R$ between their centres and charged uniformly with total charge $+Q$ and $-Q$ respectively. A particle of mass m carrying charge $+q$ is released from the centre of the ring A. If the particle reaches the centre of ring B with velocity $\left(X\left(\frac{Qq}{4\pi\varepsilon_0 Rm}\right)\right)^{1/2}$, then X is ________________.

17. Suppose we have an unlimited number of identical bulbs. Each bulb consumes power 4 W if it is operated at a potential difference 40 V. We also have a battery of emf 60 V and internal resistance $8\ \Omega$. The minimum number of bulbs that must be connected in parallel across this battery such that each bulb consumes power less than 4 W is ___________.

18. An insulating sphere of radius R is fixed with its centre at the origin of coordinates and charged such that the region $r < \frac{R}{2}$ (r denotes the distance from the centre of the sphere) has uniform charge density 2ρ and the region $\frac{R}{2} \le r \le R$ has uniform charge density ρ. A small dipole of dipole moment p is initially located at the point $A\left(\frac{3R}{2}, 0\right)$ on the X-axis with its moment vector pointing in the +X direction. If the work done in slowly taking the dipole to the point $B\left(0, -\frac{3R}{2}\right)$ along a circular path centred at the origin, such that the dipole moment vector always points in the +X direction, is $\frac{1}{N}\left(\frac{p\rho R}{\varepsilon_0}\right)$, then N is _________.



SPACE FOR ROUGH WORK

SUBJECT II : CHEMISTRY 66 MARKS

SECTION 1

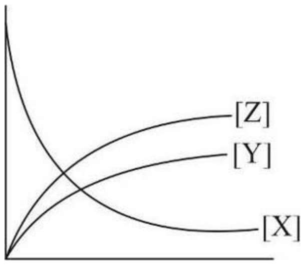
SINGLE CORRECT ANSWERS TYPE

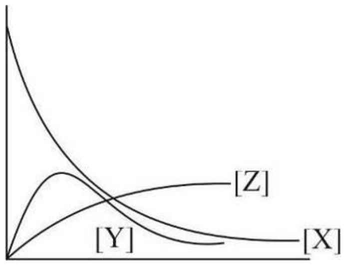
This section contains 06 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

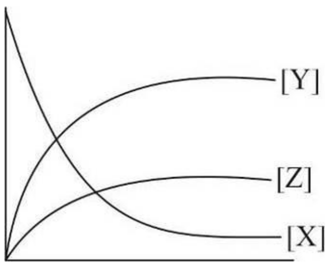
1. Which of the following concentration profile is valid for the given sequential chemical reactions ?

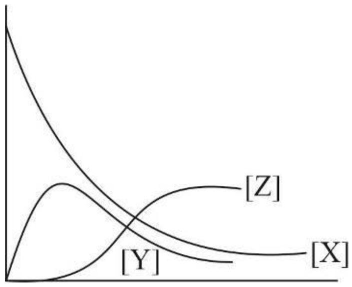
$$X \xrightarrow{k_1} Y \xrightarrow{k_2} Z$$

$k_1 = 0.5 \min^{-1}$; $k_2 = 0.05 \min^{-1}$

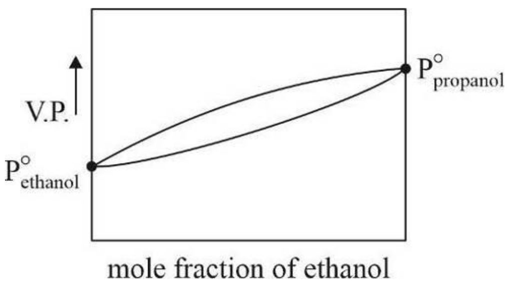
(A) 

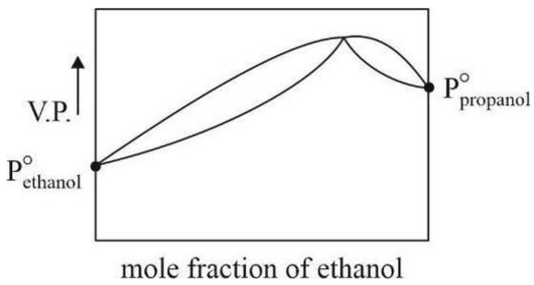
(B) 

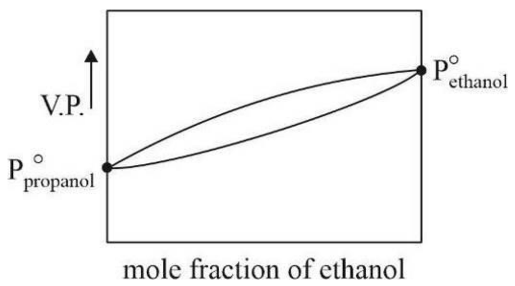
(C) 

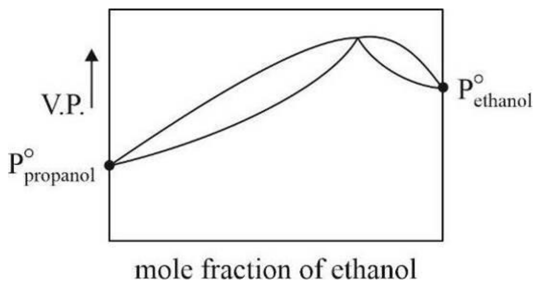
(D) 

2. Which of the following figure correctly represent fractional distillation of a mixture of ethanol and propanol ?

(A) 

(B) 

(C) 

(D) 

3. Which of the following is NOT correctly matched ?

(A) Buckminster fullerene (C-60 molecule) $\rightarrow$ Truncated icosahedron

(B) B-12 molecule $\rightarrow$ Icosahedron

(C) ZnS-cubic $\rightarrow$ Zinc blende arrangement

(D) Diamond-cubic $\rightarrow$ B.C.C arrangement

4. Gas phase thermal decomposition of di-tert-butyl peroxide is obeying first order kinetics.

$$(CH_3)_3C-O-O-C(CH_3)_3(g) \longrightarrow 2(CH_3)_2C=O(g)+C_2H_6(g)$$

Total pressure at completion of the reaction is 360 torr and it is 240 torr after 10 minutes from start of the reaction, then rate of disappearance of $(CH_3)_3C-O-O-C(CH_3)_3$ after 30 min is : [in torr / min]

(A) 1.04 **(B)** 2.87 **(C)** 6.93×10^{-2} **(D)** 0.14

5. Vapour pressure of pure water at 25°C is 80 torr and that of an aqueous solution is 75 torr at the same temperature. If K_f for water is $1.86\,K\,kg\,mol^{-1}$ then freezing point of the solution is :

(A) $-6.48°C$ **(B)** $-6.88°C$ **(C)** $-6.99°C$ **(D)** $-6.27°C$

6. What would be percentage packing efficiency of the smaller cubes, formed within a F.C.C arrangement by joining centres of the tetrahedral voids ?

[Take $(0.225)^3=0.0114$, $(0.414)^3=0.0710$, $(0.732)^3=0.392$, $(1.414)^3=2.83$, $\pi/3=1.05$]

(A) 12.22% **(B)** 48.02% **(C)** 58.07% **(D)** 23.58%

SPACE FOR ROUGH WORK

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of 06 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

7. $A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B \underset{k_4}{\overset{k_3}{\rightleftharpoons}} C$ at 25°C

$k_1=2.5\,min^{-1}$, $k_2=5.5\,min^{-1}$, $k_3=10.8\,min^{-1}$ and $k_4=4.7\,min^{-1}$

The Arrhenius factor is $10^{12}\,min^{-1}$ for each reaction then :

(A) $A \longrightarrow B$ is an endothermic process

(B) $B \longrightarrow C$ is an endothermic process

(C) Activation energy for the conversion $B \longrightarrow A$ is higher than the activation energy for the conversion $B \longrightarrow C$

(D) Overall order of the reaction is 1

8. Select the correct statements regarding colligative properties of ideal aqueous solutions :

(A) Colligative properties are driven by entropy factor

(B) Colligative properties increases with mole fraction of water

(C) Colligative properties are independent of amount of the solution

(D) Colligative properties are independent of degree on ionization of the solute dissolved in water

9. Which of the following is/are correct for a F.C.C unit cell ?

(A) It has effectively four constituent particle per unit cell

(B) It has effectively three voids per particle per unit cell

(C) Packing efficiency is 0.74

(D) It has tetrahedral, octahedral and cubic voids

10. Acid catalysed hydrolysis of (+) sucrose is a first order reaction. It gives a mixture of (+) glucose and $(-)$ fructose as the product of hydrolysis. If angle of rotation of the solution for a plane polarised light is $+32°$ at $t = 0$ and it is $-12°$ at $t = \infty$ and $+20°$ at $t = 50$ minutes then :

[Take $\log_{10} 2 = 0.30$, $\log_{10} 5 = 0.70$, $\log_{10} 11 = 1.05$, $\log_{10} 3 = 0.48$]

(A) Half life of the reaction is 100 minutes

(B) The reaction involve inversion of optical nature of the solution

(C) Solution will be optically inactive after 190 minutes

(D) After first half life solution is dextrorotatory

11. Boiling point of an aqueous solution of benzoic acid is 101°C. A solution of equal molality was prepared by dissolving benzoic acid in benzene. Then identify correct statements from the following:

[Given that for water $K_b = 0.83k\ \text{kg mol}^{-1}$, $K_f = 1.86k\ \text{kg mol}^{-1}$ and for benzene $K_b = 2.0k\ \text{kg mol}^{-1}$ and boiling point is 75°C]

(A) Freezing point of the aqueous solution of benzoic acid is $-2.24°C$

(B) Boiling point of the solution of benzoic acid in benzene is 77.4°C

(C) Observed molar mass of benzoic acid in water is equal to its value in presence of benzene as the solvent

(D) Observed molar mass of benzoic acid in water is less than to its value in presence of benzene as the solvent

12. INCORRECT statements regarding defects in crystalline solids are :

(A) Frenkel and Schottky defect are stoichiometric defects

(B) Metal excess defect always cause decrease in density of the ionic solid and increase in its electrical conductivity

(C) Metal deficiency defects always cause increase in density and electrical conductivity of the ionic solids

(D) Impurity defects creates cationic vacancies in the ionic solids

SPACE FOR ROUGH WORK

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has 06 Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*.** **(Example: 6, 81, 1.50, 3.25)**

13. Thermal decomposition of gaseous HI obey following rate law

$$2HI(g) \xrightarrow{k} H_2(g) + I_2(g)$$

$$\text{Rate} = k\,P_{HI(g)}^2$$

If the reaction starts with $P_{HI(g)} = 150\,\text{torr}$ then total pressure at half-life of the reaction will be : [in torr]

14. Boiling point of a liquid X is 127°C at 1 atm pressure. Enthalpy of vapourisation of the liquid is $40\,\text{kJ/mole}$. On addition of 50 grams of a non volatile and non dissociative solid Y to 1 L of the liquid X (density = 2 g/ml), the solution obtained will boil at T°C. Value of T is :

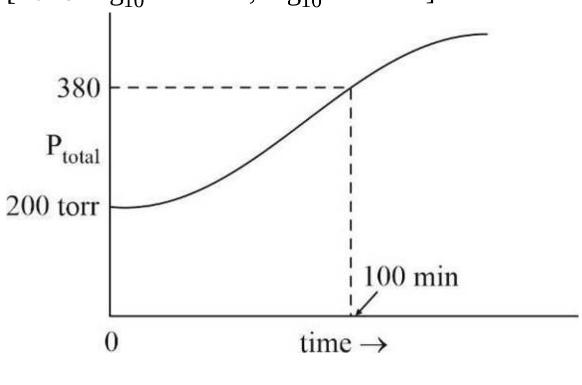
[Take : Universal gas constant $R = 8\,\text{J/mol Kelvin}$; Molar mass of $X_{(\ell)} = 100\,\text{g mol}^{-1}, Y_{(s)} = 80\,\text{g mol}^{-1}$]

15. In diamond cubic structure, effective number of constituent per unit cell are p and number of voids per unit cell occupied by carbon atoms are q. Find value of $(p \times q)$.

16. First order gas phase thermal decomposition of N_2O_5 occur as follows :

$$N_2O_5(g) \longrightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$$

If total pressure varies with time as given in the figure then half life of the reaction is : (in minutes)
[Take $\log_{10} 2 = 0.30, \log_{10} 5 = 0.70$]



17. 0.2 M NaOH was gradually added to 200 mL, 0.1 M HCl solution. After complete neutralisation of the HCl, addition of NaOH was stopped and 1.7 L of water was further added to dilute the final mixture. What would be osmotic pressure (in atm) of the diluted mixture at 35°C.
[Given that degree of dissociation of NaCl is 98% in water] [Take : $R = 0.0821\,\text{atm-L/mol-L}$]

18. Titanium oxide having formula $Ti_{2x}O_3[x < 1]$ has metal deficiency defect due to which it consist of two different oxidation states of the metal i.e. Ti^{3+} and Ti^{4+}. If ratio of number of the metal ions i.e. $Ti^{+4} : Ti^{+3}$ is 1 : 24 then value of 19x is :

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SUBJECT III : MATHEMATICS 66 MARKS

SECTION 1
SINGLE CORRECT ANSWERS TYPE

This section contains 06 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. If a function $f:[-2, \infty) \to R$ is such that $f(x) = x^2 + 4x - \left|x^2 - 4\right|$, then range of $f(x)$ is:

(A) $(-\infty, \infty)$ **(B)** $[-6, 12]$ **(C)** $[-6, \infty)$ **(D)** $[-6, 4]$

2. If m is the slope of a line which is tangent to $y^3 = x^4$ & a normal to $x^2 - 2x + y^2 = 0$, then $\left(\frac{3m}{4}\right)^3$ is equal to $(m \neq 0)$

(A) 3 **(B)** $\frac{4}{3}$ **(C)** 4 **(D)** $\frac{3}{4}$

3. Let $f:[0, \infty) \to [2, \infty)$ be a differentiable increasing and onto function satisfying $f^2(x) - 2f(x) - \sqrt{x} = f^2(y) - 2f(y) - \sqrt{y}$. The value of $\lim\limits_{x \to 0^+} \frac{f(x) - 2}{\sqrt{x}}$ is equal to:

(A) $\frac{1}{\sqrt{2}}$ **(B)** $\frac{1}{2}$ **(C)** $\frac{1}{3}$ **(D)** 0

4. Let $f:[0, \infty) \to [2, \infty)$ be a differentiable increasing and onto function satisfying $f^2(x) - 2f(x) - \sqrt{x} = f^2(y) - 2f(y) - \sqrt{y}$. If $g(x)$ be the inverse of $f(x)$, then $g'(4)$ is equal to:

(A) 16 **(B)** $\frac{1}{16}$ **(C)** 96 **(D)** $\frac{1}{96}$

5. If $P(x) = a_0 + a_1x^2 + a_2x^4 + ... + a_nx^{2n}$ be a polynomial in $x \in R$ with $0 < a_1 < a_2 < .... < a_n$, then $P(x)$ has:

(A) exactly one maximum **(B)** exactly one minimum
(C) one maximum and one minimum **(D)** None of these

6. The normal to the curve $2x^2 + y^2 = 12$ at the point (2, 2) cuts the curve again at:

(A) $\left(-\frac{22}{9}, -\frac{2}{9}\right)$ **(B)** $\left(\frac{22}{9}, \frac{2}{9}\right)$ **(C)** $(-2, -2)$ **(D)** None of these

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SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of 06 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

7. For the function $f(x)=\sqrt{4x^2-4x+2}+|x|$ which of the following holds good:

(A) The least value of the function is $\dfrac{\sqrt{3}+1}{2\sqrt{2}}$

(B) The least possible value of function is $\dfrac{\sqrt{3}+1}{\sqrt{2}}$

(C) The function takes its least value at $x=\dfrac{3-\sqrt{3}}{6}$

(D) Function has exactly one point of minima

8. If complete set of values of 'a' for which $f(x)=\log_a(4ax-x^2)$ is strictly increasing in $\left[\dfrac{3}{2}, 2\right]$, then possible value(s) of a:

(A) $\dfrac{5}{8}$ **(B)** $\dfrac{1}{4}$ **(C)** 2 **(D)** 3

9. Let $f(x)=(x-1)^p\cdot(x-2)^q$ where $p>1, q>1$. Each critical point of $f(x)$ is a point of extremum when:

(A) $p=3, q=4$ **(B)** $p=4, q=2$ **(C)** $p=2, q=3$ **(D)** $p=2, q=4$

10. If $f(x)=\left[\tan^2\left(x-\dfrac{\pi}{4}\right)\right]$ where $[x]$ is the greatest integer function, then:

(A) $f(x)$ is continuous at $x=0$

(B) $f(x)$ is differentiable at $x=\dfrac{\pi}{2}$

(C) $f(x)$ is continuous in $\left(0, \dfrac{\pi}{2}\right)$

(D) $f(x)$ is differentiable in $\left(0, \dfrac{\pi}{2}\right)$

11. If $\displaystyle\int \frac{\sin x-\cos x}{(\sin x+\cos x)\sqrt{\sin x\cos x+\sin^2 x\cos^2 x}}dx=\operatorname{cosec}^{-1}\left(g(x)\right)+c \;\forall\; x\in R$, then:

(A) $g(x)=1+\sin 2x$

(B) $g(x)=1-\sin 2x$

(C) $g(x)\ge 0$

(D) $-1\le g(x)\le 1$

12. If Rolle's theorem is applicable to the function f defined by $f(x)=\begin{cases} ax^2+b, & |x|<1 \\ 1, & |x|=1 \\ \dfrac{\lambda}{|x|} & |x|>1 \end{cases}$ $for\ x\in[-2, 2]$, then:

(A) $a+b=1$ **(B)** $\lambda=a+b$ **(C)** $b=\dfrac{3}{2}$ **(D)** $3a+b=0$

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SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has 06 Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*****. (Example: 6, 81, 1.50, 3.25)**

13. Find a, b, c such that $\lim\limits_{x\to 0}\dfrac{axe^x - b\log(1+x) + c\cdot x\cdot e^{-x}}{x^2\cdot \sin x} = 2$. The value $\dfrac{a+4b}{c}$ is _____.

14. Value of $\lim\limits_{x\to 0}\left(\dfrac{\sin x}{x}\right)^{\frac{\sin x}{x-\sin x}}$ is n. The value of $[n^{-3}]$ is ____. {where [.] denotes greatest integer function}

15. If $f(2x+1) = 4x^2 + 14x$, then find the sum of the squares of roots of the equation $f(x) = 0$.

16. Find the value of ab if $y = a\log|x| + bx^2 + x$ has its extreme values at $x = -1$ and $x = 2$

17. For how many integral values of k where $k \in (-15, 15)$ does the function $y = x^3 - 3(7-k)x^2 - 3(9-k^2)x + 2,\ x > 0$ have a point of maximum?

18. If $f(x)$ is twice differentiable and $f''(0) = p$, then $\lim\limits_{x\to 0}\dfrac{2f(x) - 3f(2x) + f(4x)}{x^2}$ is kp. The value of k is ______.

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